

Linear Algebra Cheat Sheet

Reference: Split_20260814_1325.pdf

1 Vector Spaces

A vector space $V(F)$ consists of a non-empty set V and a field F that satisfy:

- $(V, +)$ is an Abelian group.
- $\alpha(x + y) = \alpha x + \alpha y$
- $(\alpha + \beta)x = \alpha x + \beta x$
- $\alpha(\beta x) = (\alpha\beta)x$
- $1 \cdot x = x$

Properties: $\alpha \cdot 0 = 0, 0 \cdot x = 0$. If $\alpha x = 0 \implies \alpha = 0$ or $x = 0$.

2 Vector Subspaces

A non-empty subset $W \subseteq V$ is a vector subspace if it is a vector space over F with respect to the operations in V .

- **Test:** $x, y \in W; \alpha, \beta \in F \implies \alpha x + \beta y \in W$.
- Arbitrary intersection of a family of subspaces is a subspace.
- Union of two subspaces is a subspace $\iff W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

Linear Sum & Direct Sum

- $W_1 + W_2 = \{x_1 + x_2 \mid x_1 \in W_1, x_2 \in W_2\}$
- $V = W_1 \oplus W_2 \iff V = W_1 + W_2$ and $W_1 \cap W_2 = \{0\}$.
In a direct sum, elements are uniquely expressed as $x = x_1 + x_2$.
- $\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2)$

3 Linear Independence & Basis

Linear Dependence (L.D.): $\exists$ scalars $\alpha_1, \dots, \alpha_n \in F$ not all zero such that $\alpha_1 x_1 + \dots + \alpha_n x_n = 0$.

Linear Independence (L.I.): $\alpha_1 x_1 + \dots + \alpha_n x_n = 0 \implies \alpha_i = 0 \forall i$.

Basis & Dimension: A set S is a basis if S is L.I. and $[S] = V$ (generates V).

- $\dim V =$ Number of elements in a basis.
- Any set of $n + 1$ vectors in an n -dimensional space is L.D.
- For a finite field K containing q elements and $\dim V = n$, total elements $= q^n$.

4 Linear Transformations (L.T.)

$T : V_1 \rightarrow V_2$ is a linear transformation if $T(\alpha x + \beta y) = \alpha T(x) + \beta T(y)$.

- **Isomorphism:** A linear transformation that is bijective (1-1 and onto).
- **Image (Im T):** Subspace of V_2 containing $\{y : \exists x \in V_1, T(x) = y\}$.
- **Kernel (Ker T):** Subspace $\{x \in V_1 \mid T(x) = 0_2\}$.
- T is 1-1 $\iff \text{Ker } T = \{0_1\}$.
- T is Non-singular $\iff T$ is 1-1 and onto $\iff \text{Rank } T = \dim V$.

Rank-Nullity Theorem:

$\dim(\text{Ker } T) + \dim(\text{Im } T) = \dim V_1 \implies \text{Nullity}(T) + \text{Rank}(T) = \dim V_1$.

5 Matrix Properties

- **Symmetric:** $A' = A$. Dimension of subspace is $n(n+1)/2$.
- **Skew-Symmetric:** $A' = -A$. Diagonal elements are 0. Determinant of odd order is 0. $\dim$ is $n(n-1)/2$.
- **Orthogonal:** $AA' = A'A = I$. Determinant is ± 1 .
- **Hermitian:** $A^* = A$. Principal diagonals are strictly real.
- **Skew-Hermitian:** $A^* = -A$. Diagonals are 0 or purely imaginary.
- **Unitary:** $AA^* = A^*A = I$.
- **Idempotent:** $A^2 = A$.
- **Involutory:** $A^2 = I$.
- **Nilpotent:** $A^m = 0, A^{m-1} \neq 0$ (index m).

6 Rank of a Matrix

- For an $m \times n$ matrix A , $\text{Rank}(A) \leq \min(m, n)$.
- If A is an $n \times n$ non-singular matrix ($|A| \neq 0$), then $\text{Rank}(A) = n$.
- If $|A| = 0$, $\text{Rank}(A) \leq n - 1$.
- Row-Rank = Column-Rank = $\text{Rank}(A)$.

7 System of Linear Equations

Non-Homogeneous ($AX = B$):

- **Consistent:** $\rho([A : B]) = \rho(A)$.
- **Unique Solution:** $\rho([A : B]) = \rho(A) = n$.
- **Infinite Solutions:** $\rho([A : B]) = \rho(A) < n$.
- **Inconsistent:** $\rho([A : B]) \neq \rho(A)$.

Homogeneous ($AX = 0$):

- Always consistent (0 is trivial solution).
- **Non-trivial solution:** Exists if $\rho(A) < n$ (for square matrix, $|A| = 0$).
- **Dimension of solution space:** $n - \text{Rank}(A)$.

8 Eigenvalues & Eigenvectors

For square matrix A , $\lambda \in F$ is an eigenvalue if $AX = \lambda X$ for non-zero eigenvector X .

- **Characteristic Equation:** $|A - \lambda I| = 0$.
- Sum of eigenvalues = $\text{Trace}(A)$ = sum of diagonal elements.
- Product of eigenvalues = $|A|$.
- Eigenvalues of A^n are λ^n . Eigenvalues of A^{-1} are λ^{-1} .
- **Cayley-Hamilton Theorem:** Every square matrix satisfies its own characteristic equation.
- **Diagonalization:** Matrix A is diagonalizable if $B = P^{-1}AP$ is a diagonal matrix. Similar matrices have the same characteristic roots.
- **Minimal Polynomial:** Unique monic polynomial of smallest degree such that $p(A) = 0$. Minimal polynomial divides the characteristic polynomial.